Databases

Introduction to Database Systems

- Definition of database system
- Database + DBMS
- Examples: banking, e-commerce, student systems

Components of a Database System

- Data
- DBMS software
- Users (end users, DBA, developers)
- Hardware
- Procedures

Benefits of DBMS

- Data consistency
- Security & access control
- Data sharing
- Backup & recovery

Database System Architecture

- Structure of DBMS
- Data storage, processing, access
- Ensures scalability & security

Architecture Types

- One-tier
- Two-tier (client-server)
- Three-tier (presentation, application, database)

One-Tier Architecture

- All components on one system
- Simple development/testing
- Limited scalability

Two-Tier Architecture

- Client sends queries
- Server processes database
- Used in small–medium apps

Three-Tier Architecture

- UI layer
- Application logic layer
- Database layer
- Used in web systems

Data Models Overview

- Blueprint of database
- Defines structure & relationships

Types of Data Models

- Conceptual (ER)
- Logical (tables & relations)
- Physical (storage details)

Common Data Models

- Hierarchical
- Network
- Relational
- Object-Oriented

Relational Database Basics

- Tables, rows, columns
- Relationships among tables

Keys in Databases

- Primary key
- Foreign key
- Candidate key
- Composite key

Relationships

- One-to-One
- One-to-Many
- Many-to-Many

Relational Algebra vs Calculus

- Algebra: procedural
- Calculus: non-procedural
- Foundation of SQL

Relational Algebra Operations

- Selection
- Projection
- Union
- Difference
- Cartesian product

SQL Overview

- Standard language for relational DB
- Used to store, retrieve, update data

SQL Command Types

- DDL: CREATE, ALTER
- DML: INSERT, UPDATE, DELETE
- DQL: SELECT
- DCL: GRANT, REVOKE

Basic SQL Example

- CREATE TABLE
- INSERT data
- SELECT queries

Advanced SQL

- Joins
- Subqueries
- Aggregate functions
- Views & indexes

Database Design

- Reduce redundancy
- Improve integrity
- Efficient queries

Normalization

- 1NF, 2NF, 3NF
- Organize tables effectively

Transactions

- Logical unit of work
- COMMIT and ROLLBACK

Concurrency Control

- Manage simultaneous access
- Locking
- Timestamp methods

Recovery Management

- Logs
- Checkpoints
- Restore after failures

Indexing & File Organization

- Heap files
- Sequential files
- Indexes improve search

Query Optimization

- Execution plans
- Join strategies
- Reduce I/O cost

Security & Modern Databases

- Authentication & authorization
- Distributed DBMS
- NoSQL basics
- Data warehousing